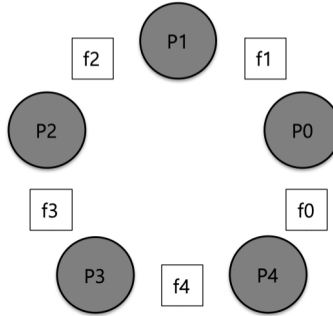


Name: _____

Assume I have 5 philosophers (P0 to P5) and 5 forks (F0 to F5). Each philosopher needs repeatedly think and eat. To eat, the philosopher needs both forks, to the left and to the right.



The default approach is to grab the fork on the left, followed by the fork on the right, using `wait(fork)` which waits until the fork is acquired. Explain how you would change this approach to prevent deadlocks, breaking each of the conditions listed.

1. Prevent **mutual exclusion** : threads claim exclusive control of resources that they acquire.
2. Prevent **hold and wait** : threads hold resources allocated to them while waiting for additional resources
3. Prevent **no preemption** : resources cannot be forcibly removed from threads that are holding them
4. Prevent **circular wait** : there exists a circular chain of threads such that each thread holds one or more resources that are being requested by the next thread in the chain.